

## UEN008: ENERGY AND ENVIRONMENT

| L | T | P | Cr  |
|---|---|---|-----|
| 2 | 0 | 0 | 2.0 |

**Course Objective:** The exposure to this course would facilitate the students in understanding the terms, definitions and scope of environmental and energy issues pertaining to current global scenario; understanding the value of regional and global natural and energy resources; and emphasize on need for conservation of energy and environment.

### Syllabus

**Introduction:** Natural Resources & amp; its types, Concept of sustainability and sustainable use of natural resources, Pollution based environmental issues and case-studies.

**Conventions on Climate Change:** Origin of Conference of Parties (COPs), United Nations Framework Convention on Climate Change (UNFCCC) and Intergovernmental Panel on Climate Change (IPCC); Kyoto Protocol, instruments of protocol – CDM, JI and IET; Montreal Action Plan; Paris Agreement and post-Paris scenario.

**Air Pollution:** Origin, Sources and effects of air pollution; Primary and secondary meteorological parameters; Wind roses; Atmospheric Stability; Inversion; Plume behavior; Management of air pollution: Source reduction and Air Pollution Control Devices for particulates and gaseous pollutants in stationary and mobile sources.

**Water Pollution:** Origin, Sources of water pollution, Category of water pollutants, Physico-Chemical characteristics, Components of wastewater treatment systems, Advanced treatment technologies.

**Solid Waste Management:** Introduction to solid waste management, Sources, characteristics of municipal and industrial solid waste, Solid waste management methods: Incineration, composting, Biomethanation, landfill, E-waste management, Basal convention.

**Energy Resources:** Classification of Energy Resources; Conventional energy, resources- Coal, petroleum and natural gas, nuclear energy, hydroelectric power; Non-conventional energy resources – Biomass energy, Thermo-chemical conversion and biochemical conversion route; Generation of Biogas and biodiesel as fuels; Solar energy-active and passive solar energy absorption systems; Type of collectors; Thermal and photo conversion applications; Wind energy.

**Facilitated through Online Platforms:**

*Approved in 1st meeting of the Academic Council held on September 11, 2025. Further revision approved in 3rd meeting of the Academic Council held on March 18, 2026, incorporating modification in credits due to the addition of AI courses.*

**Ecology and Environment:** Concept of an ecosystem; structural and functional units of an ecosystem; Food Chain, Food Web, Trophic Structures and Pyramids; Energy flow; Ecological Succession; Types, Characteristics, Biodiversity, Biopiracy.

**Human Population and the Environment:** Population growth, variation among nations; Population explosion – Family Welfare Programmes; Environment and human health; Human Rights; Value Education; Women and Child Welfare; Role of Information Technology in Environment and Human Health, Environmental Ethics.

### **Laboratory Work (if applicable)**

### **Course Learning Objectives (CLO)**

The students will be able to:

1. Comprehend the interdisciplinary context with reference to the environmental issues and case studies.
2. Assess the impact of anthropogenic activities on the various elements of environment and apply suitable techniques to mitigate their impact.
3. Conceptualize and explain the structural and functional features of ecological systems.
4. Correlate environmental concerns with the conventional energy sources associated and assess the uses and limitations of non-conventional energy technologies.

### **Text Books**

1. Moaveni, S., Energy, Environment and Sustainability, Cengage (2018)
2. Down to Earth, Environment Reader for Universities, CSE Publication (2018)
3. Chapman, J.L. and Reiss, M.J., Ecology - Principles and Application, Cambridge University Press (LPE) (1999).
4. Eastop, T.P. and Croft, D.R., Energy Efficiency for Engineers and Technologists, Longman and Harlow (2006).
5. O'Callagan, P.W., Energy Management, McGraw Hill Book Co. Ltd. (1993).
6. Peavy H.S. and Rowe D.R. Environmental Engineering, McGraw Hill (2013).

### **Evaluation Scheme:**

| <b>Sr. No.</b> | <b>Evaluation Elements</b>              | <b>Weightage (%)</b> |
|----------------|-----------------------------------------|----------------------|
| 1.             | MST                                     | 25-35                |
| 2.             | EST                                     | 35-45                |
| 3.             | Sessionals/Quizzes/Projects Evaluations | 30                   |

*Approved in 1st meeting of the Academic Council held on September 11, 2025. Further revision approved in 3rd meeting of the Academic Council held on March 18, 2026, incorporating modification in credits due to the addition of AI courses.*